Karan Yadav

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EDUCATION

Netaji Subhas University of Technology, New Delhi **2023 – 2027**
Bachelor of Technology, ME *CGPA: 6.91/10*

Gyan Sagar Public School, Dwarka, New Delhi *Class XII: 91% | Class X: 94%*

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript, TypeScript, Bash **Analytics:** Pandas, NumPy, scikit-learn, Advanced Excel, Power BI, Tableau
Data Engineering & Cloud: ETL/data pipelines, data quality; Azure Data Factory, Azure Data Lake, Databricks, Snowflake, dbt, GCP (Cloud Functions, Cloud Tasks), AWS (EC2), Docker, GitHub Actions
AI & LLMs: RAG, LLM tool-calling, token optimization, OpenAI APIs, LangChain, MLflow **Core:** FastAPI, React, Node.js, Flutter, Asyncio, DSA, OOP, System Design, REST APIs, Pytest

EXPERIENCE

Full Stack Developer Intern **Jun – Jul 2026**
Hyper Pixel Games, New Delhi

- Architected an async backend pipeline on GCP using Cloud Functions and Cloud Tasks with a transactional outbox, guaranteeing **zero lost jobs** under high upload concurrency.
- Fixed container OOM crashes during heavy multi-image uploads by streaming files directly to Cloud Storage via signed URLs, and implemented rate limits and abuse quotas.
- Offloaded compute-heavy image cropping and compression to background Dart Isolates, preventing main UI thread freezing to maintain responsive user interactions.

SELECTED PROJECTS

Interactive Analytics Dashboards | Tableau Public & Power BI *Tableau, Power BI, Python/Pandas*
github.com/karnyadavdev/startup-funding-dashboard

- **Startup Funding Landscape (Tableau):** cleaned a 5,006-row deal dataset to **3,918 verified rounds** (2015–2025) with Python/Pandas; built a 6-sheet interactive dashboard — 2021 peaked at **$13.2B** / 539 deals; top-3 sectors hold ~50%; early-stage share fell **13 points** by 2025.
- **Retail Profitability (Power BI):** analyzed **9,994 transactions** ($2.3M sales, **12.5%** margin) with KPIs, regional analysis and discount diagnosis — every discount tier above **20%** runs at a loss (**–$135K** vs **+$422K**); Furniture: 32% of sales, 6% of profit.

AskKara | Live Web Research & Knowledge Retrieval Engine *React, TypeScript, FastAPI, Python, Asyncio, RAG, NDJSON*
d24884eoij7cl7.cloudfront.net | github.com/karnyadavdev/askKara

- Built an autonomous research engine using FastAPI and LLM tool-calling, decomposing complex queries into parallel sub-searches for multi-hop retrieval across live, client-rendered web applications.
- Integrated custom YADA microservice to fetch up to 10 live web sources concurrently using bounded asyncio workers, cutting total query latency by **40%**.
- Built a custom DOM sanitizer removing scripts, tracking pixels, and ads, converting raw HTML to clean markdown to slash LLM token consumption by **90%+**; results stream to a React UI over an NDJSON stream.

YADA | High-Speed Web Extraction Engine for RAG *Python, FastAPI, curl_cffi, Asyncio, Playwright, Docker, AWS EC2*
github.com/karnyadavdev/YADA-Scraper-API

- Built an async extraction API with FastAPI, cascading from TLS-impersonated HTTP (curl_cffi) to headless Playwright browser execution to scrape protected, client-rendered SPAs at sub-second speeds.
- Built crash-safe domain memory with atomic disk writes, 3-strike failure handling, and deferred batching, achieving a **4.7x speedup** and **0% data loss** across 200 tasks.
- Engineered stealth anti-bot evasion (Cloudflare, Akamai, DataDome) via TLS fingerprint impersonation and Playwright warm-session reuse; enforced async DNS SSRF validation and an 8 MB stream cap, containerized on AWS EC2, verified with **198 automated tests**.

LEADERSHIP, TEAMWORK & COMMUNICATION

- Ranked **3rd across NSUT** in GDSC Competitive Programming Team selection; mentored 20+ students in weekly DSA sessions.
- Organized **Code Conundrum 2024**, an ICPC-style Codeforces contest with 100+ participants.
- Competitive Programming: LeetCode: 1600 Rating (**Top 25%**) | Codeforces: 1100+ Rating | Chess: 1800+ Peak Rapid.